

DBMS Practical Assignment

1. Create the following tables (primary keys are underlined.)

Property(pno,description, area)

Owner(oname,address,phone)

An owner can have one or more properties, but a property belongs to exactly one owner. Create the relations accordingly, so that the relationship is handled properly and the relations are in normalized form (3NF).

- a) Insert 5 records into owner table.
- b) insert 5 property records for each owner .
- c) Update phone no of “Mr. Patil” to 9800566330
- d) Delete all properties from “pune” owned by “ Mr. Nene”

2. Create the following tables (primary keys are underlined.).

Emp(eno,ename,designation,sal)

Dept(dno,dname,dloc)

There exists a one-to-many relationship between emp & dept.

Create the Relations accordingly, so that the relationship is handled properly and the relations are in normalized form (3NF).

- a) Insert 5 records into department table
- b) Insert 5 employee records for each department.
- c) increase salary of “managers” by 15%;
- d)change location of department 2 to ‘Mumbai’

3. Consider the following database

Person (pnumber, pname, birthdate, income)

Area (aname, area_type,pincode)

The relationship is as follows:Person-Area:many-to-one.

The ‘area_type’ can have values ‘Urban’ and ‘Rural’ Create normalized tables (3NF) for the above and solve the following queries.

Write & execute insert/ update / delete statements for following business tasks

- a) Insert 5 records in the Person table.
- b) Insert 3 records in the Area table
- c) List the names of all people income less than 100000
- d) Change the pincode of ‘kalyaninagar’ to 411036.

4. Consider the following database

Doctor (dno, dname, city)

Patient (pno, pat_name, city, disease)

The relationship is as follows: Doctor-Patient: many-to-many with Fee as a descriptive attribute Create normalized tables (3NF) for the above and solve the following queries.

Write & execute insert/ update / delete statements for following business tasks

- a) Insert 5 records in the Doctor table.
- b) Insert 5 records in the Patient table.
- c) List the details of all the doctors from _____ city.
- d) List the names of the patients suffering from ‘cancer’ disease.

5. Consider the following database

Room (room_no,room_name,room_type,charges)

Guest(Guest_code, Gname,city)

The relationship is as follows: Room-Guest: one-to-one. room_type can have values as either 'AC' or 'NonAC'. Create normalized tables (3NF) for the above and solve the following queries.

Write & execute insert/ update / delete statements for following business tasks

- a) Insert 5 records in the Room table.
- b) Insert 5 records in the Guest table.
- c) Find the maximum charges of a room.
- d) List the names of all the NONAC rooms whose charges are more than 10000.

6. Consider the following database

Employee(eno,ename,designation,salary)

Department(dno,dname,location)

The relationship is as follows: Employee-Department: many-to-one.

- a) Insert 5 records in the Employee table.
- b) Insert 5 records in the Department table.
- c) Display average salary of an employee
- d) Display the details of employees whose names ends with an alphabet ____.

7. Consider the following database

Employee(eno,ename,designation,salary)

Department(dno,dname,location)

The relationship is as follows: Employee-Department: many-to-one.

- a) Insert 5 records in the Employee table.
- b) Insert 5 records in the Department table.
- c) Count the number of employees in dept no ____.
- d) Display the details of employees whose names starts with ____.

8. Create the following tables (primary keys are underlined).

Account (acct_no ,acct_type, balance, branch_name)

Customer (cust_no, cust_name, cust_city)

Relationships : Customer-Account :1-M Constraints: acct_type can be "saving" or "current", balance > 0 Execute the following queries

- a) Insert 5 records in the Account table.
- b) Insert 5 records in the Customer table.
- c) Count customers in city "_____".
- d) Find the total balance at branch "M.G.Road".

9. Create the following tables (primary keys are underlined).

Account (acct_no ,acct_type, balance, branch_name)

Customer (cust_no, cust_name, cust_city)

Relationships : Customer-Account :1-M Constraints: acct_type can be "saving" or "current", balance > 0 Execute the following queries

- a) Insert 5 records in the Account table.
- b) Insert 5 records in the Customer table.
- c) Give the total balance of all Saving accounts.
- d) Find the maximum balance of an account.

10. Consider the relations

Person (pnumber, pname, birthdate, income),

Area(aname,area_type).

An area can have one or more person living in it , but a person belongs to exactly one area.

The attribute 'area_type' can have values as either urban or rural.

Create the Relations accordingly, so that the relationship is handled properly and the relations are in normalized form (3NF).

Write select queries for following business tasks and execute them.

- a) Insert 5 records in the Person table.
- b) Insert 5 records in the Area table.
- c) List details of all people whose names start with the alphabet ' _ ' & contains maximum _ alphabets in it.
- d) List the names of all people whose birthday falls in the month of .

11. Consider the relations

Person (pnumber, pname, birthdate, income),

Area(aname,area_type).

An area can have one or more person living in it , but a person belongs to exactly one area. The attribute 'area_type' can have values as either urban or rural.

Create the Relations accordingly, so that the relationship is handled properly and the relations are in normalized form (3NF).

Write select queries for following business tasks and execute them.

- a) Insert 5 records in the Person table.
- b) Insert 5 records in the Area table.
- c) Give the count of people who are born on ' '
- d) List the sum of incomes of people living in ' '

12. Create the following relations, for an investment firm

emp(emp-id ,emp-name, address, date)

Investor(inv-name , inv-no, inv-date, inv-amt)

An employee may invest in one or more investments, hence he can be an investor. But an investor need not be an employee of the firm.

Create the Relations accordingly, so that the relationship is handled properly and the relations are in normalized form (3NF).

Assume appropriate data types for the attributes. Add any new attributes , as required by the queries. Insert sufficient number of records in the relations / tables with appropriate values as suggested by some of the queries.

Write the following queries & execute them.

- a) Insert 5 records in the emp table.
- b) Insert 5 records in the Invertor table.
- c) Find the maximum inv-amt of an Investor.
- d) List the sum of inv-amt of an Investor.

13. Create the following tables (primary keys are underlined.).

Emp(eno,ename,designation,sal)

Dept(dno,dname,dloc)

There exists a one-to-many relationship between emp & dept.

Create the Relations accordingly, so that the relationship is handled properly and the relations are in normalized form (3NF).

- a) Insert 5 records into department table
- b) Insert 5 employee records for each department.
- c) Delete all employees of deparment no 5;
- e) Change location of department 2 to 'Mumbai'

14. Consider the following database

Person (pnumber, pname, birthdate, income)

Area (aname, area_type,pincode)

The relationship is as follows: Person-Area: many-to-one.

The 'area_type' can have values 'Urban' and 'Rural'. Create normalized tables (3NF) for the above and solve the following queries.

Write & execute insert/ update / delete statements for following business tasks

- a) Insert 5 records in the Person table.
- b) Insert 3 records in the Area table
- d) Change the pincode of 'kalyaninagar' to 411036.
- e) Delete all area where the area_type is 'Rural'

15. Consider the following Student-Teacher database

Student (sno ,s_name, s_class) s_class can be either "FY", "SY" or "TY"

Teacher (tno ,t_name, yrs_experience)

The relationship is as follows: Student-Teacher: M-M with descriptive attribute Subject.

Create normalized tables (3NF) for the above and solve the following queries.

Write & execute insert/ update / delete statements for following business tasks

- a) Insert 5 records in the Student table.
- b) Insert 5 records in the Teacher table
- c) Delete all students who are in class "FY"
- e) Update the class as 'TY' whose name 'Ramesh'